SeongRae Noh

rhosunr99@korea.ac.kr | seongraenoh.github.io | linkedin.com/in/seongrae-noh | github.com/SeongRaeNoh

Research Summary

Integrated MS-Ph.D. student building LLM agents that turn open-ended instructions into verifiable action sequences in 3D environments — spanning scene editing, multi-agent cooperation under partial observability, and camera control.

Technical Skills

Agentic AI & LLMs: LLM Agents, Multi-Agent Systems, Agentic Planning, LLM Evaluation, HuggingFace Transformers
ML & Embodied AI: PyTorch, Vision-Language Models, Embodied Task Planning, VirtualHome, Unreal Engine 5
Tools & Languages: Python, C++, Docker, Git; Korean (Native), English (Advanced), Japanese (Intermediate)

Research Experience

Graduate Researcher — Mar. 2024 – Present
Korea University, IIIXR Lab (Advisor: Prof. HyeongYeop Kang) — *Seoul, South Korea*

- Conducting research on LLM agents, embodied AI, multimodal planning, and agent evaluation in interactive 3D environments.
- Developed agentic planning methods for open-vocabulary 3D scene editing and uncertainty-aware multi-agent decision-making.

Undergraduate Researcher — Dec. 2021 – Feb. 2024
KyungHee University, IIIXR Lab (Advisor: Prof. HyeongYeop Kang) — *Yongin, South Korea*

- Designed and implemented an embodied LLM agent framework in VirtualHome, enabling multi-agent task planning, communication, and action validation for long-horizon household tasks.
- Developed generative 3D environment pipelines for agent simulation, creating urban layouts and Unreal Engine scenes for interactive embodied AI experiments.

Industrial Projects

Development of a Multimodal Information-Driven Service Agent Framework (LG Electronics) — Oct. 2025 – Sep. 2026

- Designed a multimodal agent architecture for HVAC service workflows, combining LLM reasoning, visual understanding, and structured device data for context-aware diagnosis and action recommendation.

Understanding Multimodal PDF Documents Using LLMs (LG Electronics) — Jun. 2025 – Sep. 2025

- Built an LVLM-based document intelligence system for multimodal technical PDFs, enabling structured extraction, table/figure understanding, and cross-modal reasoning over service manuals.

Selected Publications

(SIGGRAPH Asia 2026, Conditionally Accepted) CuCaT: Cut-Structured Camera Trajectory Generation for Dance Music Videos
SooBin Lim*, **SeongRae Noh***, Jeongyoun Kwon, YoungBeom Kim, HyeongYeop Kang†

- Proposed a cut-structured camera formulation that separates cut and transition events over a performer-relative shot plan, achieving the best continuity and variation on the DCM benchmark while keeping the dancer visible in nearly every frame.

(CVPR 2026) Edit-As-Act: Goal-Regressive Planning for Open-Vocabulary 3D Indoor Scene Editing
SeongRae Noh, SeungWon Seo, Gyeong-Moon Park†, HyeongYeop Kang†

- Formulated open-vocabulary 3D scene editing as agentic goal-regressive planning over world states, and introduced the E2A Benchmark, improving instruction fidelity by 20% and semantic consistency by 43.1%.

(ICLR 2026) From Assumptions to Actions: Turning LLM Reasoning into Uncertainty-Aware Planning for Embodied Agents
SeungWon Seo*, SooBin Lim*, **SeongRae Noh***, Haneul Kim, HyeongYeop Kang†

- Proposed uncertainty-aware scenario tree planning for LLM agents under partial observability, converting latent assumptions into robust multi-agent decisions while reducing communication by 71.1% and improving task performance by 8.6%.

(AAAI 2025 Oral) REVECA: Adaptive Planning and Trajectory-Based Validation in Cooperative Language Agents Using Information Relevance and Relative Proximity
SeungWon Seo*, **SeongRae Noh***, Junhyeok Lee, SooBin Lim, Won Hee Lee, HyeongYeop Kang†

- Developed a cooperative LLM-agent planning framework with relevance-filtered memory and trajectory-based validation, improving performance by 47% in noisy environments and user preference by 73%.

Education

Korea University — Seoul, South Korea
Integrated MS-Ph.D. in Computer Science, IIIXR LAB (Advisor: Prof. HyeongYeop Kang) — *Mar. 2024 – Feb. 2029*

KyungHee University — Yongin, South Korea
B.S. in Computer Science — *Mar. 2020 – Feb. 2024*

Honors & Service

Awards: AAAI Student Scholarship (2025); Teaching Assistance Scholarship (2024); Pearl Abyss Fellowship (2023, 2021); 2nd place, KHU Software Convergence Inter-Club Competition (2022)
Service: Reviewer – NeurIPS, ICML (2026), ACM MM, Scientific Reports (2025); Student Volunteer – AAAI (2025)